



DECUS

PROGRAM LIBRARY

DECUS NO.	FOCAL8-70
TITLE	ANALYSIS OF VARIANCE RANDOMIZED BLOCK "F" TEST
AUTHOR	C. T. Lund
COMPANY	Canada Department of Agriculture Ontario, Canada
DATE	October 22, 1969
SOURCE LANGUAGE	FOCAL

ATTENTION

This is a USER program. Other than requiring that it conform to submittal and review standards, no quality control has been imposed upon this program by DECUS.

The DECUS Program Library is a clearing house only; it does not generate or test programs. No warranty, express or implied, is made by the contributor, Digital Equipment Computer Users Society or Digital Equipment Corporation as to the accuracy or functioning of the program or related material, and no responsibility is assumed by these parties in connection therewith.

ANALYSIS OF VARIANCE RANDOMIZED BLOCK "F" TEST

DECUS Program Library Write-up

DECUS No. FOCAL8-70

FOR EACH ANALYSIS

A Analysis of Variance Table

I Degrees of Freedom:

- 1) Between Replicates
- 2) Between treatments
- 3) Error
- 4) Total

II Sum of Squares:

- 1) Between Replicates
- 2) Between treatments
- 3) Error
- 4) Total

III Mean Squares:

- 1) Between Replicates
- 2) Between treatments
- 3) Error

IV "F" Value:

- 1) Between Replicates
- 2) Between treatments

B Grand Mean

Restrictions

The block design must be complete.

1) In a 4K memory, up to a 10×10 block may be analyzed if the FOCAL extended functions are retained. A 30×30 block was analyzed without an error recovery when the extended functions were deleted.

MATHEMATICAL METHOD USED

- a = Number of Replicates
- y = Number of treatments
- x = Individual observation
- N = Total number of observations
- T = Sum of values in treatment
- R = Sum of values in Replicate

$$\text{Sum of Squares for treatments } \sum \frac{T^2}{a} - \frac{(\sum x)^2}{N}$$

$$\text{Sum of Squares for Replicates } \sum \frac{R^2}{y} - \frac{(\sum x)^2}{N}$$

$$\text{Total Sum of Squares } \sum x^2 - \frac{(\sum x)^2}{N}$$

Error Sum of Squares = SS total - (SS treatment and SS Replicates)

Mean squares are found by dividing the sum of squares by the appropriate number of degrees of freedom (a-1), (y-1) and (N-1). The "F" values are the ratios of the mean squares to the mean square error.

W
C-FOCAL., 1968

```

Ø1.Ø2 T  !!, "ANALYSIS OF VARIANCE RANDOMIZED BLOCK F TEST", !!
Ø1.Ø3 E
Ø1.Ø4 A  "HOW MANY REPLICATES", BL, !
Ø1.Ø5 A  "HOW MANY TREATMENTS", PO, !!!
Ø1.Ø6 T  "ENTER EACH TREATMENT BY REPLICATE", !!
Ø1.1Ø F  N=1,1,PO;T !;F B=1,1,BL;A A;G 1.3
Ø1.3Ø S  Y(N)=Y(N)+A;S YS=YS+A↑2;S D(B)=D(B)+A
Ø1.31 F  N=1,1,PO;S EX=EX+Y(N)↑2;S SX=SX+Y(N)
Ø1.61 F  N=1,1,BL;S EY=EY+D(N)↑2
Ø1.64 S  C=SX↑2/PO*BL;S ST=(EY/PO)-C;S SP=(EX/BL)-C
Ø1.65 S  SS=(YS)-C;S SE=SS-(ST+SP);S DT=ST/(BL-1)
Ø1.66 S  DP=SP/(PO-1)
Ø1.67 S  ER=SE/(PO-1)*(BL-1);S BF=DT/ER;S TF=DP/ER
Ø1.68 T  !!! "ANALYSIS OF VARIANCE TABLE", !!!
Ø1.7Ø T  "SOURCE      DF      SUM OF SQUARES      MEAN SQUARES"
Ø1.71 T  "      F", !!!
Ø1.8Ø T  "BET REP", %5, BL-1, "      ", %, ST, "      ", DT, "      "
Ø1.81 T  %4.Ø2, BF, !!
Ø1.9Ø T  "BET TRE", %5, PO-1, "      ", %, SP, "      ", DP, "      "
Ø1.91 T  %4.Ø2, TF, !!

Ø2.1Ø T  "ERROR  ", %5, (BL-1)*(PO-1), "      ", %, SE, "      ", ER, !!
Ø2.2Ø T  "TOTAL  ", %5, BL*PO-1, "      ", %, SS, !!
Ø2.3Ø T  "GRAND MEAN", SX/BL*PO, !!!, "END OF JOB", !!!;G 1.Ø3
*
```

G

ANALYSIS OF VARIANCE RANDOMIZED BLOCK F TEST

HOW MANY REPLICATES:5

HOW MANY TREATMENTS:4

ENTER EACH TREATMENT BY REPLICATE

:7 :4 :5 :9 :7

:3 :6 :2 :4 :8

:6 :4 :5 :8 :7

:9 :6 :8 :3 :5

ANALYSIS OF VARIANCE TABLE

SOURCE	DF	SUM OF SQUARES	MEAN SQUARES	F
BET REP=+	4	=+0.970007E+01	=+0.242502E+01	=+ 0.47
BET TRE=+	3	=+0.100000E+02	=+0.333333E+01	=+ 0.65
ERROR =+	12	=+0.615000E+02	=+0.512500E+01	
TOTAL=+	19	=+0.812001E+02		

GRAND MEAN=+0.580000E+01

END OF JOB

HOW MANY REPLICATES: ?01.00

*